

Fig. 3: SD/MMC memory card pins

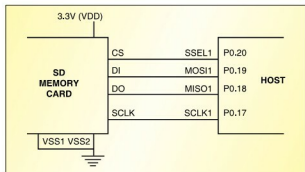


Fig. 5: Interfacing of SD memory card to the host

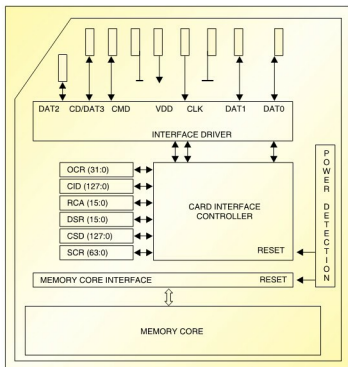


Fig. 6: SD memory card architecture

Signals and Functions of SD Card Pins in SD and SPI Modes

Pin	Signal	Function (SD mode)	Function (SPI mode)
1	DAT3/CS	Data line 3	Chip select/slave select (SS)
2	CMD/DI	Command line	Master-out slave-in (MOSI)
3	VSS1	Ground	Ground
4	VDD	Supply voltage	Supply voltage
5	CLK	Clock	Clock (SCLK)
6	VSS2	Ground	Ground
7	DAT0/DO	Data line 0	Master-in slave-out (MISO)
8	DAT1/IRQ	Data line 1	Unused or IRQ
9	DAT2/NC	Data line 2	Unused

interfaces to the host point-to-point (in Fig. 5, an ARM microcontroller is the host). This type of interfacing is very popular in the industry.

In serial peripheral interface (SPI) mode, you can use following signals of the host:

1. CS: Host to card chip-select signal
2. CLK: Host to card clock signal
3. MOSI (master-out slave-in): Host to card single-bit data signal
4. MISO (master-in slave-out): Card to host single-bit data signal

Now many companies are manufacturing suitable hosts for the SD bus interface. For example, Philips is manufacturing LPC2148 microcontroller with MOSI and MISO

signals.

Master-slave mode of communication is used for multiple slave devices in the SD architecture. MOSI is a unidirectional signal used to transfer serial data from the master to the slave. When the host is master, data can move from the host to the SD card. That's why MOSI is connected to data input (DI) of the SD/MMC card. The MISO signal transfers serial data from the slave to the master. When the SD is a slave, serial data is output on MISO signal. When the SD is a master, it clocks in serial data from this signal.

SD memory cards use 1- or 4-bit bus width and star topology to connect multiple cards, while MMC cards use 1-bit bus width and bus topology for reading multiple cards.

Fig. 6 shows the internal architecture of an SD memory card. There are various components in the card such as operation conditions register, card identification number (card individual number for identification), driver stage register and relative card address (local system address of a card, dynamically suggested by the card and approved by the host during initialisation). Memory core is the actual storage of the memory. This is a very compact and delicate architecture. ●

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